

THOUGHT EXPERIMENT

Given: a universe where it is proven that $P \neq NP$ (Assumption 1)

Given: a graph G

Consider: a maximum matching M , a minimum vertex cover S

Fact 1: $|M| \leq |S|$

Fact 2: Finding M is in P

Fact 3: Finding S is **NP-complete**

World 1

- We are not given S , M .

- We do not know anything about finding S , hence we are trying to understand it better. We start by understanding its bound.

(This simulates how research on solving most open problems proceeds.)

Problem 1: Find a lower bound on the size of S for a graph G .

Solution 1: Find a maximum matching M . $|M|$ is a lower bound [Fact 1].

Conclusion 1.1: Finding a bound on the size of an **NP-complete** problem is in P .

Conclusion 1.2: Finding a bound on the size of a problem is easier than finding the exact value.

(Both conclusions seem logical.)

World 2

- We are not given M , S .

- We do not know anything about finding M , hence we are trying to understand it better. We start by understanding its bound.

(This simulates how research on solving most open problems proceeds.)

Problem 2: Find an upper bound on the size of M for a graph G .

Solution 2: Find a min vertex cover S . $|S|$ is an upper bound [Fact 1].

Conclusion 2.1: Finding a bound on the size of a problem in P is **NP-complete**.

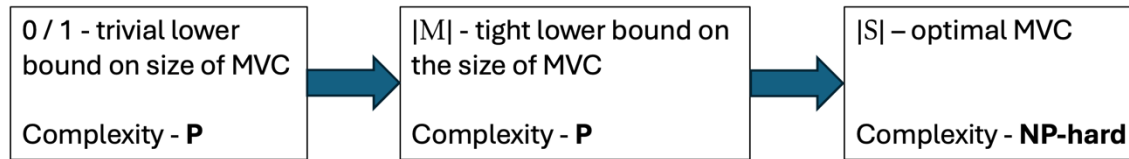
Conclusion 2.2: Finding a bound on the size of a problem is harder than finding the exact value.

(Both conclusions defy logic! Shouldn't finding a bound be at least as easy as finding the exact number, in line with most math research?)

Disclaimer: for the purposes of this thought experiment, we interchangeably use the decision version and the search version of a problem, and NP-hardness and NP-completeness.

Minimum Vertex Cover (MVC)

Smooth progression in complexity: From finding a lower bound for MVC to finding the exact size of MVC



Maximum Matching (MM)

Irregular progression in complexity: From finding an upper bound for MM to finding the exact size of MM

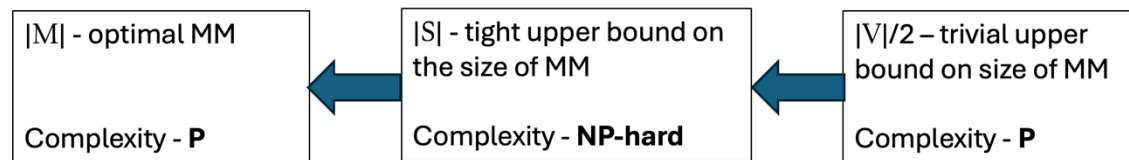


Figure 1: An illustration depicting progression of computational complexity from finding trivial bound to exact value for (top) minimum vertex cover problem and (bottom) maximum matching problem. The contradictory / anomalous nature of the progression and corresponding conclusions (1.1, 1.1, 2.1 and 2.2) seems to defy logic, especially given that one can find a matching M to make Problem 2 efficient. If $P=NP$, such anomalies do not exist.